CYCLING STORE ASSISTANT

AI CHATBOT DESIGN DOCUMENT

Natalia Ovejero López
ARTIFICIAL INTELLIGENCE

INDEX

Executive Summary

This project focuses on the design and development of an AI chatbot for an online cycling store. The main idea behind the chatbot is to make the shopping experience easier and more intuitive for users by offering quick support and guidance.

Instead of navigating through multiple pages or searching for information manually, users can simply interact with the chatbot to get what they need. The chatbot helps in three main areas: recommending bikes, answering common questions (such as shipping or returns), and assisting users in choosing the right bike size.

The chatbot was built using Voiceflow, combining structured workflows with AI-generated responses. This allows the system to provide answers that feel natural, while still keeping the conversation organized and relevant.

Additionally, the chatbot supports both English and Spanish. It automatically adapts to the language used by the user, which improves accessibility and makes the interaction more comfortable.

Overall, the goal of this project is to demonstrate how conversational AI can improve user experience in an e-commerce environment.

User Research

To design the chatbot effectively, two main user profiles were considered.

User 1: Beginner Cyclist

Carlos is a 25 year old user who wants to buy his first bike. He does not have much knowledge about cycling, so he needs simple explanations and clear guidance.

His main concerns are:

- Understanding which type of bike to choose
- Staying within a limited budget
- Avoiding technical or complex language

For users like Carlos, the chatbot needs to be friendly, easy to understand, and capable of guiding the conversation step by step.

User 2: Experienced Cyclist

Laura is a 32 year old experienced cyclist who is looking to upgrade her current bike. She already understands the basics and is more interested in performance and specific features.

She expects:

- More precise recommendations
- Less basic explanations
- Faster and more direct answers

The chatbot should be flexible enough to adapt to this type of user by providing more relevant and concise information.

Conversation Flow

The chatbot is structured around different workflows, each designed to handle a specific type of request. This helps keep the interaction organized and efficient.

At the beginning of the conversation, the chatbot greets the user and asks what they need help with. Based on the response, it directs the user to the appropriate workflow.

Bike Recommendation Workflow

If the user is looking for a bike, the chatbot asks a few key questions:

- What type of cycling they are interested in (road, mountain, or urban)
- Their budget range
- Their experience level

Once it gathers this information, it provides a recommendation that matches the user's needs. The idea is not just to give an answer, but to guide the user through a simple decision-making process.

FAQ Workflow

This workflow handles common questions about the store, such as:

- Shipping times
- Return policy
- Warranty information

For example:

- Shipping typically takes 2–4 business days
- Returns are accepted within 30 days
- Warranty depends on the product

These answers are kept short and clear to make the experience faster for the user.

Size Help Workflow

When users ask about sizing or mention their height, the chatbot provides guidance on the most appropriate bike frame size.

For example:

- A user who is 175 cm tall will usually be recommended a medium frame

This helps reduce uncertainty and improves confidence before making a purchase.

Routing Logic

One of the key aspects of the chatbot is its ability to detect user intent and route the conversation correctly.

- If the user wants to buy a bike → Bike Recommendation
- If the user asks about policies → FAQ
- If the user mentions size or height → Size Help

This ensures that each request is handled in the most appropriate way.

Sample Dialogues

Below are some examples of typical interactions.

Scenario 1 – Bike Recommendation

User: *I want a mountain bike under 800€*
Bot:
The chatbot asks a few follow-up questions about experience and terrain, and then suggests a suitable option based on those answers.

Scenario 2 – FAQ

User: *How long is shipping?*
Bot:
"Standard shipping usually takes between 2 and 4 business days."

Scenario 3 – Size Help

User: *I am 175 cm tall*
Bot:
"You will typically fit a medium frame size, although it can vary slightly depending on the brand."

AI Integration Strategy

The chatbot combines AI-generated responses with structured workflows created in Voiceflow.

The global prompt defines how the chatbot should behave, including tone, role, and general guidelines. This ensures consistency across all interactions.

Workflows are used to manage step-by-step conversations, especially in cases where the chatbot needs to collect information before giving an answer.

Finally, routing logic connects everything together, allowing the chatbot to respond appropriately depending on the user’s request.